

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

Treemap Layout Functions and Constants

Visualization Data Preparation Functions

File Data Preparation Functions

File Data Preparation

```

Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.AggregationSkippedNull} {category: UNRECOGNIZED} {title: The query contains an aggregation function that skips null values.} {description: null value eliminated in set function.} {position: None} for query: "// List git files with commit statistics\n\n MATCH (git_file:File&Git&!Repository)\n WHERE git_file.deletedAt IS NULL // filter out deleted files\n WITH percentileDisc(git_file.createdAtEpoch, 0.5) AS medianCreatedAtEpoch\n ,percentileDisc(git_file.lastModificationAtEpoch, 0.5) AS medianLastModificationAtEpoch\n ,collect(git_file) AS git_files\n UNWIND git_files AS git_file\n WITH *\n ,datetime.fromepochMillis(coalesce(git_file.createdAtEpoch, medianCreatedAtEpoch)) AS fileCreatedAtTimestamp\n ,datetime.fromepochMillis(coalesce(git_file.lastModificationAtEpoch, git_file.createdAtEpoch, medianLastModificationAtEpoch)) AS fileLastModificationAtTimestamp\n MATCH (git_repository:Git&Repository)-[:HAS_FILE]->(git_file)\n MATCH (git_commit:Git&Commit)-[:CONTAINS_CHANGE]->(git_change:Git&Change)-->(old_files_included:Git&File&!Repository)-[:HAS_NEW_NAME*0..3]->(git_file)\n RETURN git_repository.name + '/' + git_file.relativePath AS filePath\n ,split(git_commit.author, ' <')[0] AS author\n ,count(DISTINCT git_commit.sha) AS commitCount\n ,date(max(git_commit.date)) AS lastCommitDate\n ,max(date(fileCreatedAtTimestamp)) AS lastCreationDate\n ,max(date(fileLastModificationAtTimestamp)) AS lastModificationDate\n ,duration.inDays(date(max(git_commit.date)), date()).days AS daysSinceLastCommit\n ,duration.inDays(max(fileCreatedAtTimestamp), datetime()).days AS daysSinceLastCreation\n ,duration.inDays(max(fileLastModificationAtTimestamp), datetime()).days AS daysSinceLastModification\n AS maxCommitSha\n ORDER BY filePath ASCENDING, commitCount DESCENDING"

```

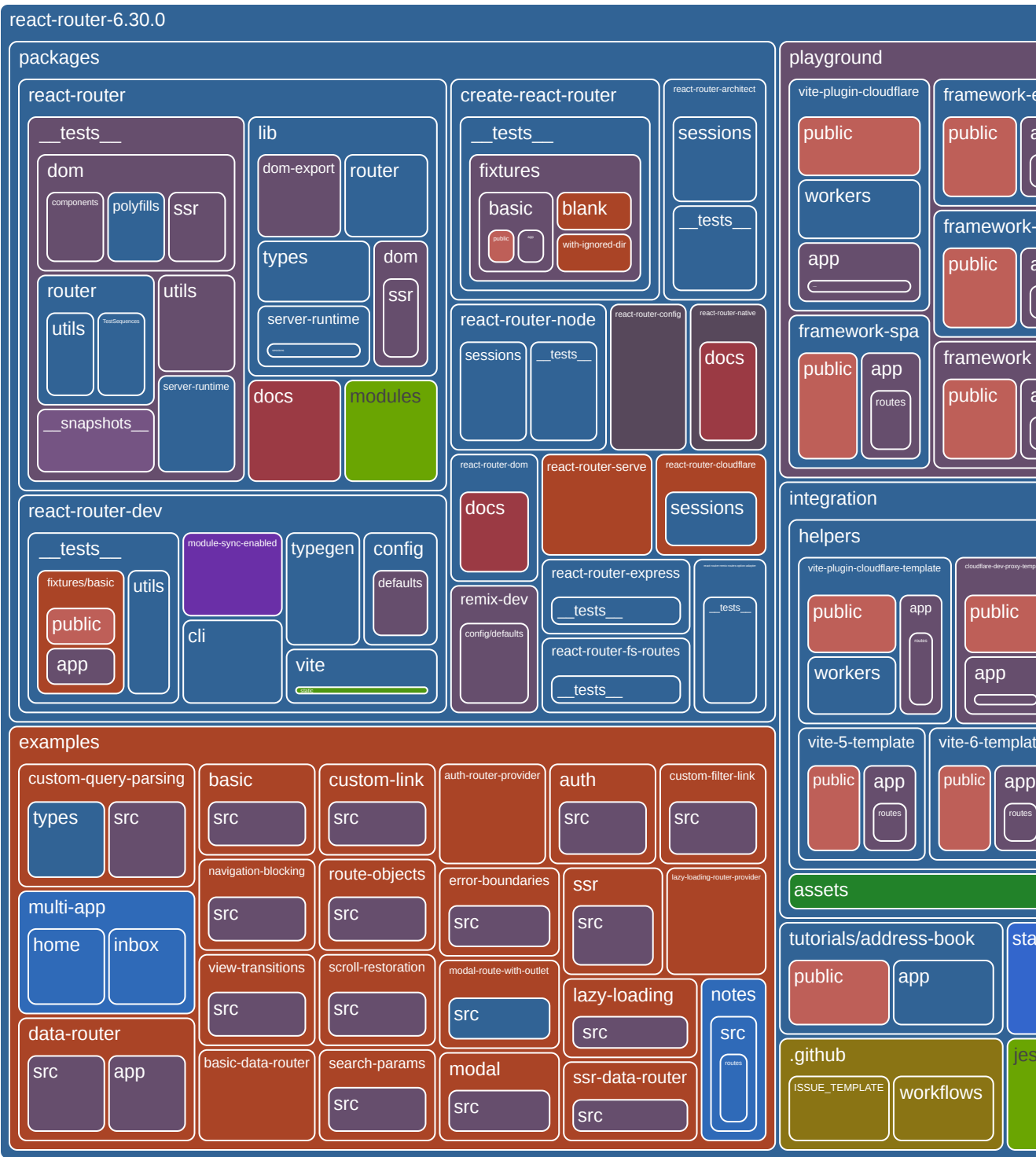
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	198.000000	198.000000	198.000000	198.000000	198.000000	198.000000
mean	22.489899	13.449495	659.520202	109.121212	392.19697	174.939394
std	100.699721	39.140447	3249.715697	221.568670	470.42799	303.195697
min	1.000000	2.000000	2.000000	2.000000	2.000000	1.000000
25%	2.000000	3.000000	22.250000	11.000000	65.000000	10.000000
50%	5.000000	6.000000	64.000000	11.000000	142.000000	25.000000
75%	13.000000	10.000000	255.500000	120.000000	616.250000	143.500000
max	1280.000000	483.000000	37505.000000	1235.000000	1754.000000	1288.000000

	directoryPath	directoryParentPath	directoryName	fileCount	firs
0	react-router-6.30.0/examples/custom-query-parsing/types	react-router-6.30.0/examples/custom-query-parsing	types	1	react-router-6.30.0/examples/custom-query-parsing/types/jsur
1	react-router-6.30.0/examples/data-router/src	react-router-6.30.0/examples/data-router	src	1	react-router-6.30.0/examples/data-router/src/la
2	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app/routes	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app	routes	1	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app/routes/_inde
3	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/public	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template	public	1	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/public/favico
4	react-router-6.30.0/integration/helpers/vite-5-template/app/routes	react-router-6.30.0/integration/helpers/vite-5-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-5-template/app/routes/_inde
5	react-router-6.30.0/integration/helpers/vite-5-template/public	react-router-6.30.0/integration/helpers/vite-5-template	public	1	react-router-6.30.0/integration/helpers/vite-5-template/public/favico
6	react-router-6.30.0/integration/helpers/vite-6-template/app/routes	react-router-6.30.0/integration/helpers/vite-6-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-6-template/app/routes/_inde
7	react-router-6.30.0/integration/helpers/vite-6-template/public	react-router-6.30.0/integration/helpers/vite-6-template	public	1	react-router-6.30.0/integration/helpers/vite-6-template/public/favico
8	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app/routes	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app/routes/_inde
9	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/public	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template	public	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/public/favico
10	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/workers	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template	workers	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/workers/a
11	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app/routes	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app	routes	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app/routes/hom
12	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/public	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic	public	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/public/favico
13	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/blank	react-router-6.30.0/packages/create-react-router/__tests__/fixtures	blank	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/blank/package
14	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/with-ignored-dir	react-router-6.30.0/packages/create-react-router/__tests__/fixtures	with-ignored-dir	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/with-ignored-dir/package
15	react-router-6.30.0/packages/react-router-architect/sessions	react-router-6.30.0/packages/react-router-architect	sessions	1	react-router-6.30.0/packages/react-router-architect/sessions/arcTableSessionStora
16	react-router-6.30.0/packages/react-router-cloudflare/sessions	react-router-6.30.0/packages/react-router-cloudflare	sessions	1	react-router-6.30.0/packages/react-router-cloudflare/sessions/workersKVStora
17	react-router-6.30.0/packages/react-router-config	react-router-6.30.0/packages	react-router-config	1	react-router-6.30.0/packages/react-router-config/LIC
18	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app/routes	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app	routes	1	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app/routes/_inde
19	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/public	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic	public	1	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/public/favico
20	react-router-6.30.0/packages/react-router-dev/vite/static	react-router-6.30.0/packages/react-router-dev/vite	static	1	react-router-6.30.0/packages/react-router-dev/vite/static/refresh-uti
21	react-router-6.30.0/packages/react-router-dom/docs	react-router-6.30.0/packages/react-router-dom	docs	1	react-router-6.30.0/packages/react-router-dom/docs/README
22	react-router-6.30.0/packages/react-router-express/__tests__	react-router-6.30.0/packages/react-router-express	__tests__	1	react-router-6.30.0/packages/react-router-express/__tests__/server-ti
23	react-router-6.30.0/packages/react-router-native/docs	react-router-6.30.0/packages/react-router-native	docs	1	react-router-6.30.0/packages/react-router-native/docs/README
24	react-router-6.30.0/packages/react-router-node/sessions	react-router-6.30.0/packages/react-router-node	sessions	1	react-router-6.30.0/packages/react-router-node/sessions/fileStora
25	react-router-6.30.0/packages/react-router/__tests__/__snapshots__	react-router-6.30.0/packages/react-router/__tests__	__snapshots__	1	react-router-6.30.0/packages/react-router/__tests__/__snapshots__/route-matc test.tsx
26	react-router-6.30.0/packages/react-router/__tests__/dom/components	react-router-6.30.0/packages/react-router/__tests__/dom	components	1	react-router-6.30.0/packages/react-router/__tests__/dom/components/LazyComponer
27	react-router-6.30.0/packages/react-router/__tests__/dom/polyfills	react-router-6.30.0/packages/react-router/__tests__/dom	polyfills	1	react-router-6.30.0/packages/react-router/__tests__/dom/polyfills/drop-Forml submit
28	react-router-6.30.0/packages/react-router/docs	react-router-6.30.0/packages/react-router	docs	1	react-router-6.30.0/packages/react-router/docs/README

Most frequent file extension per directory

Most frequent file extension per directory



Number of commits per directory

Number of git commits



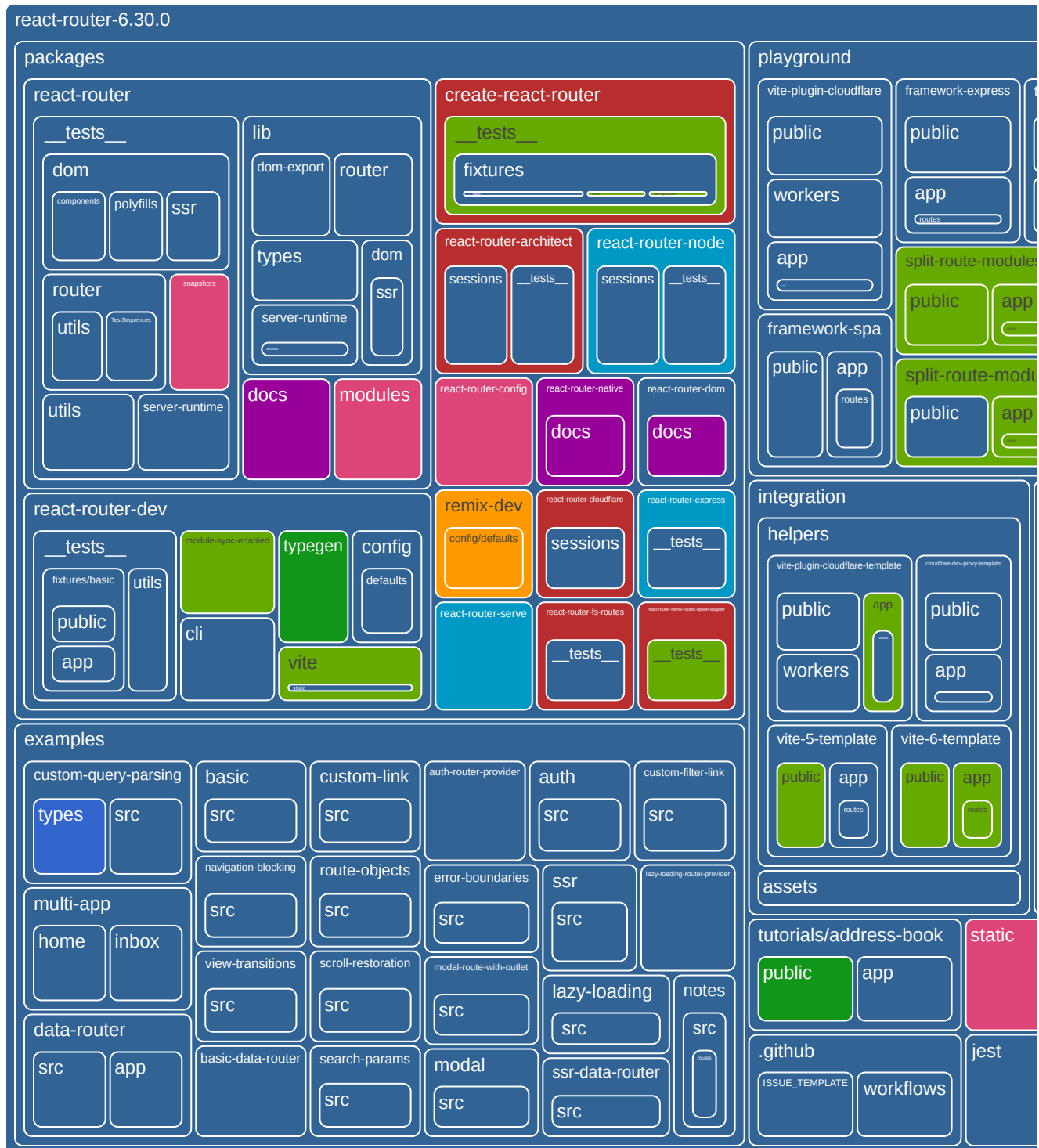
Number of distinct authors per directory

Number of distinct commit authors



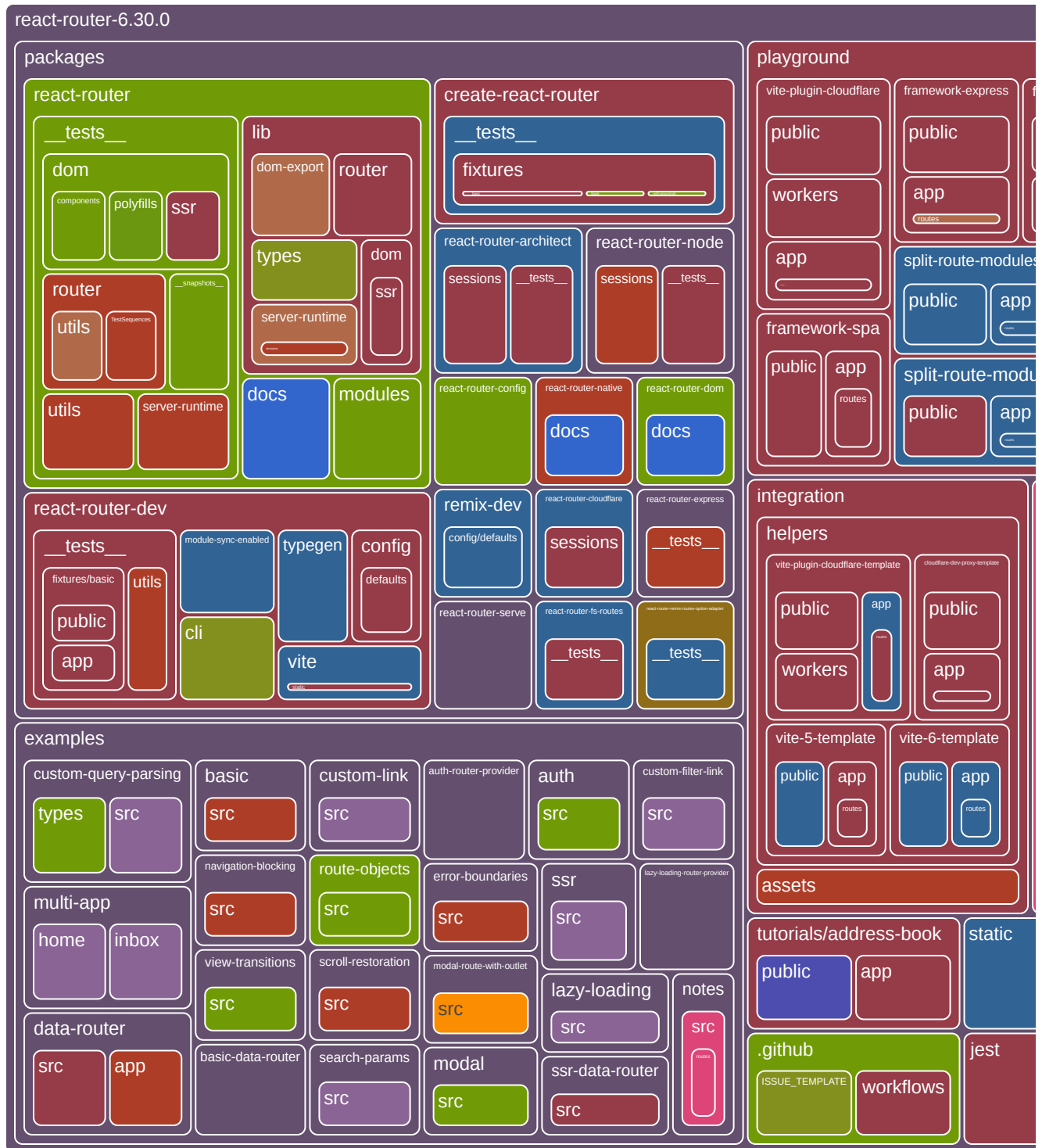
Main author per directory

Main authors with highest number of commits



Second author per directory

Second author with the second highest number of commits



Days since last commit per directory

Days since last commit



Days since last commit per directory (ranked)

Rank of days since last commit

react-router-6.30.0

packages

react-router

__tests__

dom
components polyfills ssr

router

utils

__snapshots__

lib

dom-export

router

types

server-runtime

docs

modules

create-react-router

__tests__

fixtures

basic

blank

with-ignored-dir

react-router-node

sessions

tests

react-router-dom

docs

react-router-serve

react-router-cloudflare

sessions

react-router-architect

sessions

tests

react-router-config

docs

react-router-native

playground

vite-plugin-cloudflare

public

workers

app

framework-spa

public

app

routes

integration

helpers

vite-plugin-cloudflare-template

public

app

workers

vite-5-template

public

app

routes

assets

tutorials/address-book

public

app

.github

ISSUE_TEMPLATE

workflows

react-router-dev

__tests__

fixtures/basic

public

app

utils

module-sync-enabled

cli

typegen

config

defaults

vite

static

examples

custom-query-parsing

types

src

multi-app

home

inbox

data-router

src

app

basic

src

navigation-blocking

src

view-transitions

src

basic-data-router

src

custom-link

src

route-objects

src

scroll-restoration

src

search-params

src

auth-router-provider

src

error-boundaries

src

modal-route-with-outlet

src

modal

src

auth

src

ssr

src

lazy-loading

src

ssr-data-router

src

custom-filter-link

src

lazy-loading-router-provider

src

notes

src

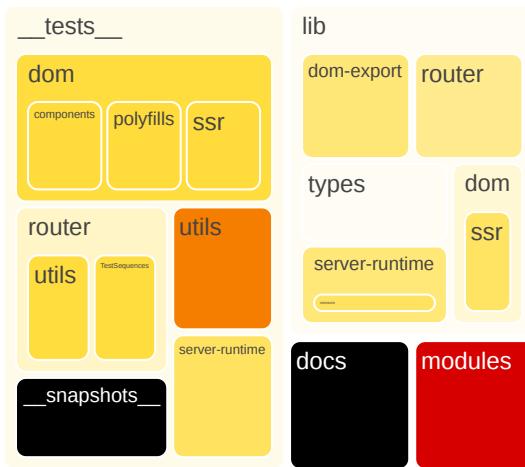
Days since last file creation per directory

Days since last file creation

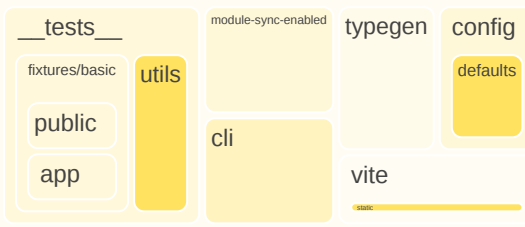
react-router-6.30.0

packages

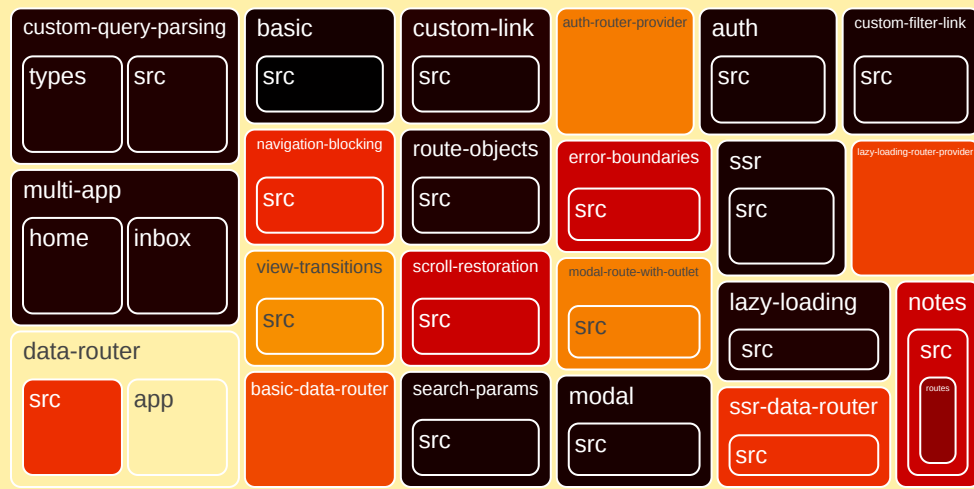
react-router



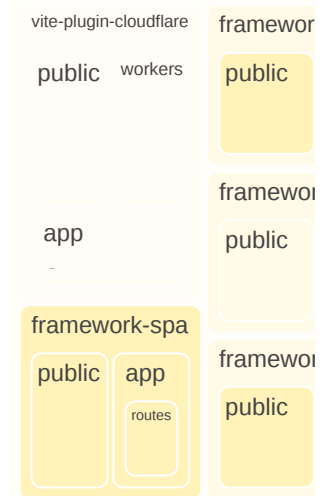
react-router-dev



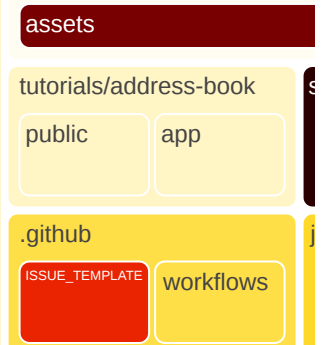
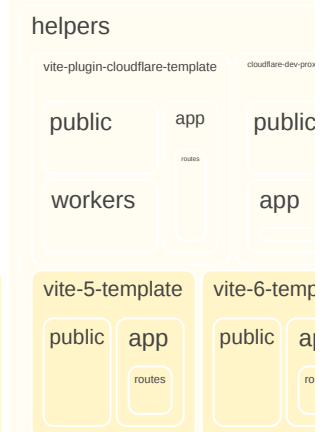
examples



playground



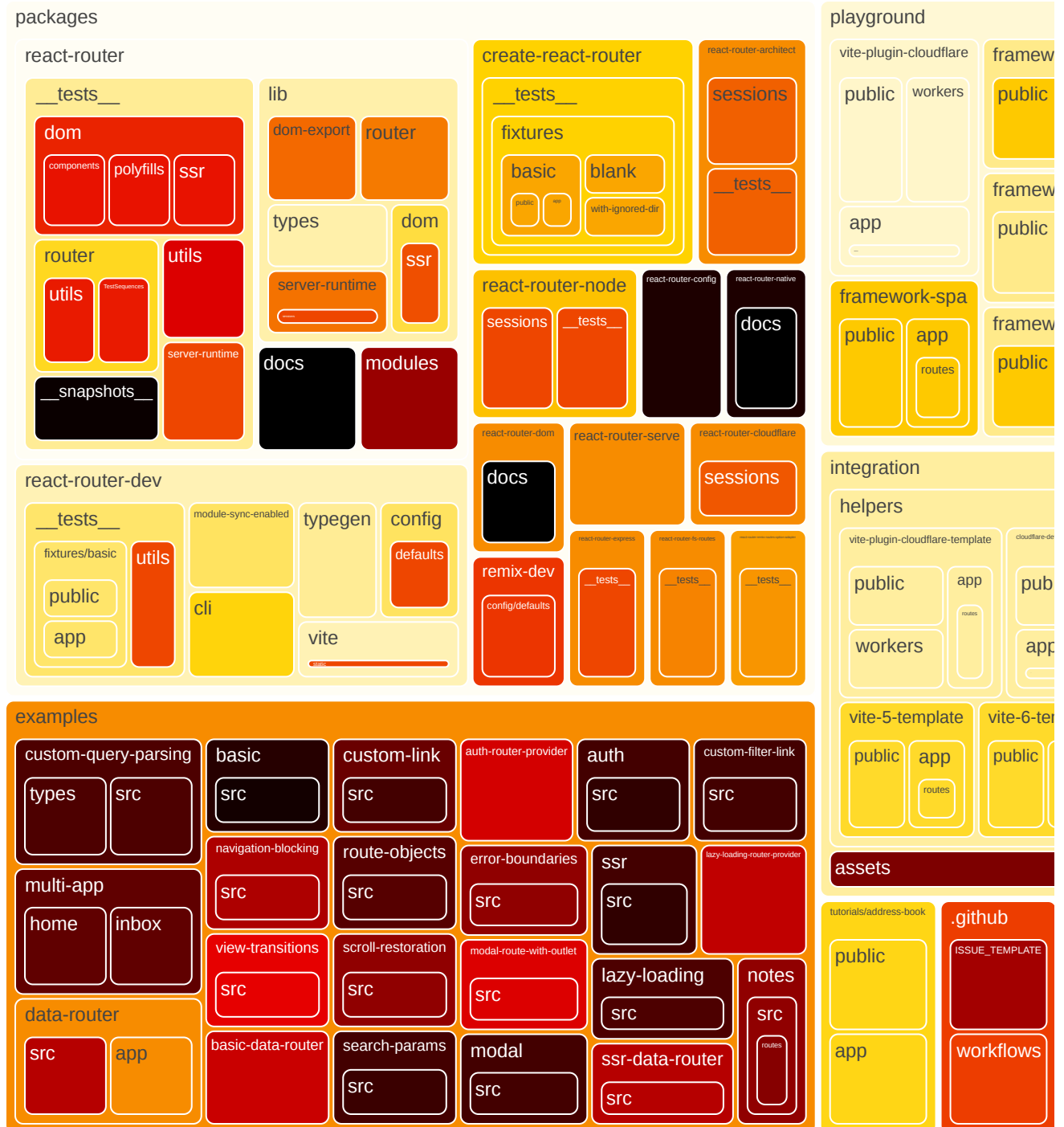
integration



Days since last file creation per directory (ranked)

Rank of days since last file creation

react-router-6.30.0



Days since last file modification per directory

Days since last file modification



Days since last file modification per directory (ranked)

Rank of days since last file modification

react-router-6.30.0



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

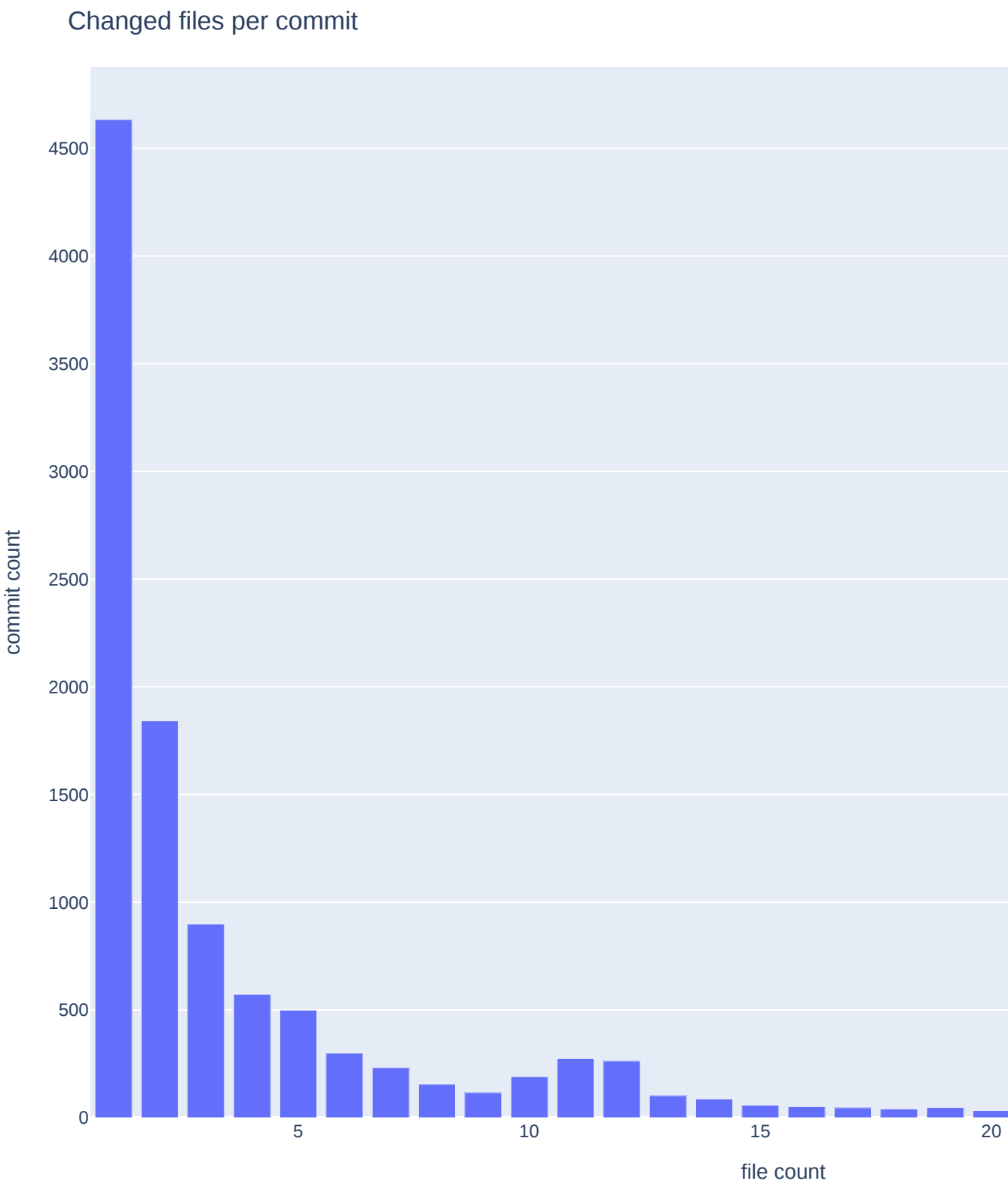
Preview data

Sum of commits that changed more than 30 files (each) = 462
Max changed files with one commit = 1061

	filesPerCommit	commitCount
count	171.000000	171.000000
mean	149.959064	65.766082
std	193.597592	390.426273
min	1.000000	1.000000
25%	43.500000	1.000000
50%	87.000000	3.000000
75%	157.500000	8.000000
max	1061.000000	4633.000000

	filesPerCommit	commitCount
0	1	4633
1	2	1842
2	3	899
3	4	573
4	5	499
5	6	300
6	7	232
7	8	155
8	9	116
9	10	190
10	11	275
11	12	263
12	13	102
13	14	87
14	15	58
15	16	51
16	17	46
17	18	40
18	19	47
19	20	33
20	21	36
21	22	31
22	23	53
23	24	30
24	25	30
25	26	15
26	27	56
27	28	56
28	29	19
29	30	17

Bar chart with the number of files per commit distribution



WordCloud of git authors

	word	frequency
0	Matt Brophy	2059
1	Michael Jackson	1951
2	Ryan Florence	1306
3	Chance Strickland	511
4	Remix Run Bot	493
5	Mark Dalglish	418
6	Pedro Cattori	412
7	Jimmy Jia	389
8	Tim Dorr	388
9	Jacob Ebey	290

Wordcloud of git authors

